COMP-4250

Assignment 1

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1. The values suitable for c depend on the value k of the hash function. Generally, for the hash function h(x) = x mod k, k is a positive integer, it is best to use c = 1in order for h(x) to range from 0 to k. In this case k would be the number of buckets available. Otherwise, the hash function will put many keys into the same bucket wasting many bucket spots. An example is if h = 10 and c = 5, then only 2 buckets are being used of the possible 10 (h(5) = 5, h(10) = 0, h(15) = 5, h(20) = 0 etc). If c = 1 instead then, h(1) = 1, h(2) = 2, h(13) = 3 etc.

2.
   a. Start
      map<key, value> << input document; // key is line number and value is int in document
      sort(map); // Sort by key
      output << reduce(map, &max()) // Reduce each same key values using max function and mapped values

      Ex: each division finds it own max, then reduce finds the max of each partition

   b. Start
      map<key, value> << input document; // key is line number and value is int in document
      sort(map); // Sort by key
      output << reduce(map, &average()) // Reduce each same key values using average function and mapped values

   c. Start
      map<key, value> << input document; // key is line number and value is int in document
      sort(map); // Sort by key

output << reduce(map, &set()) // Reduce each same key values using set function and mapped values

d. Start
map<key, value> << input document; // key is line number and value is int in document
sort(map); // Sort by key
output << reduce(map, &sum()) // Reduce each same key values using sum function and mapped values

3. Item i in basket b iff b % i == 0
   a. If s = 5, then items which b % i == 0 for when b ranges from 1 to 100 and i is the single item 5 times or more. Example: when i = 1 it appears in 100 buckets (100 > s, frequent), when i = 50 it appears in 2 buckets (2 < s, not frequent). Therefore, only items 20 and less can be frequent.
   b. For pairs only numbers 20 and under and the two items, i1 and i2, divide 5 of the same numbers in b without remainder. Example: 1 and 2 are in 50 of the same buckets because 1 appears in all and 2 appears in half.
   c. Bucket with 12 elements will be the largest such as bucket 96: {1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 96}
   d. {5, 7} -> 2
      Confidence({5, 7} -> 2) = supp({5, 7}U{2}) / supp({5, 7}) = 1 / 2
      Interest({5, 7} -> 2) = confidence({5, 7} -> 2) – supp{2} = 1/2 – 50/100 = 0
      Confidence({2,3,4} -> 5) = supp({2,3,4}U{5}) / supp({2, 3, 4}) = 1/8
      Interest({2,3,4} -> 5) = Confidence({2,3,4} -> 5) – supp({5}) = 1/8 – 1/5 = -0.075
   e. Any pair in which the second number is a multiple of the first number has %100 confidence Ex: confidence({2} -> 4) = supp({2}U{4}) / sup({4}) = 25 / 25 = %100

4. i % b = 0, 10 % 10

5. a. Every pair of items in the data is a frequent item. The lowest probability of two items is item 10 and 9 together. Each has 1/10 and 1/9 probability of occurring. Them occurring together is (1/10)*(1/9) = 0.0111111 which is still above the threshold of 1%.
   b. Interest(a -> b) = conf(a -> b) – supp(b) from our data the min supp() >= 10% maximum conf() = conf(1 -> 10)

6. support(1) = 4
   support(2) = 6
   support(3) = 7
   support(4) = 8
   support(5) = 6
   support(6) = 4
   support(1,2) = 2
   support(1,3) = 3
   support(1,4) = 2
   support(1,5) = 1
   support(1,6) = 0
   support(2,3) = 3
   support(2,4) = 4
   support(2,5) = 2
   support(2,6) = 1
   support(3,4) = 4
   support(3,5) = 4
   support(3,6) = 2
   support(4,5) = 3
   support(4,6) = 3
   support(5,6) = 2

   b. buckets containing i = 1 go to bucket j as 1*j mod 11 = j (j <=6)
   buckets containing i = 2 go to bucket 2*j except {2,6} as 2*6 mod 11 = 1
   buckets containing i = 3 repeat to buckets 3, 6, 9, 1,4, 7, 10
   buckets containing i = 4 repeat to buckets 4, 8, 1, 5, 9, 2
   buckets containing i = 5 repeat to buckets 5, 10, 4, 9, 3, 8
   buckets containing I = 6 repeat to buckets 6, 1, 7, 2

   c. buckets 8, 1, 4

   d. pairs (2,4), (3,4), (3,5)